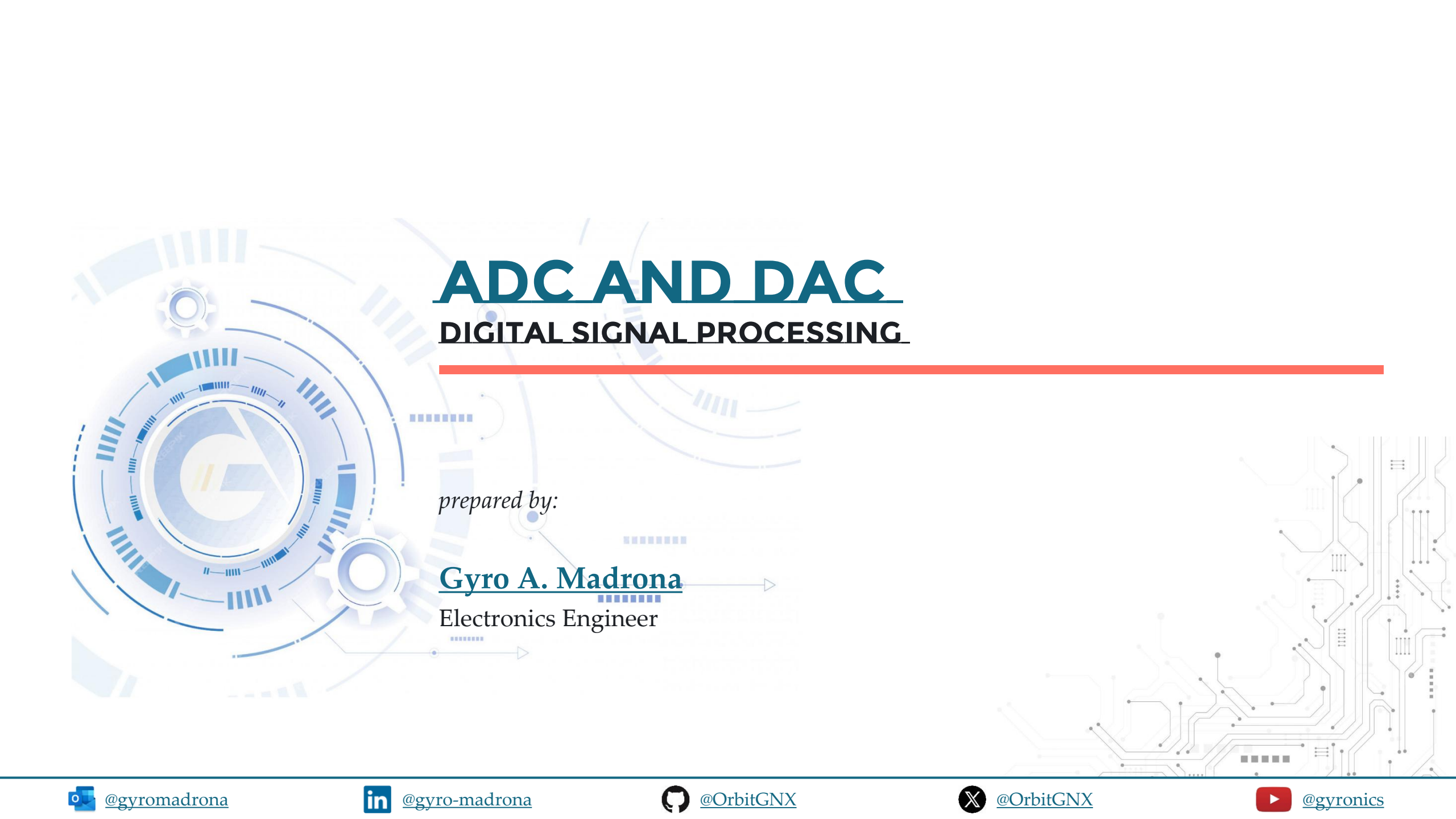




ADC AND DAC

DIGITAL SIGNAL PROCESSING

prepared by:

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TOPIC OUTLINE

Analog-to-Digital Conversion

Digital-to-Analog Conversion

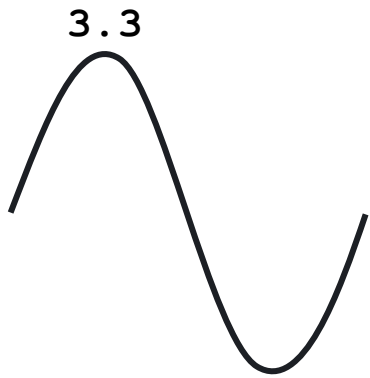


ADC

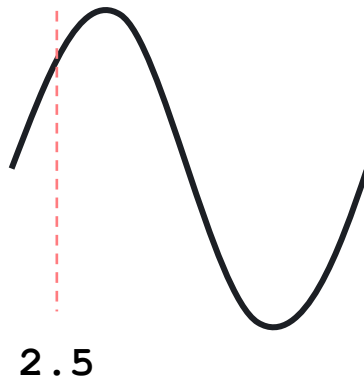


ADC PROCESS

Analog Voltage



Sampling



Quantization

A 12-bit ADC can represent $2^{12} = 4096$ different levels, ranging from 0 to 4095.

$$\text{ADC} = \frac{v_{in}}{v_{ref}} \cdot (2^n - 1)$$

$$\text{ADC} = \frac{2.5}{3.3} \cdot 4095$$

Digital Value

$$\text{ADC} = 3102$$

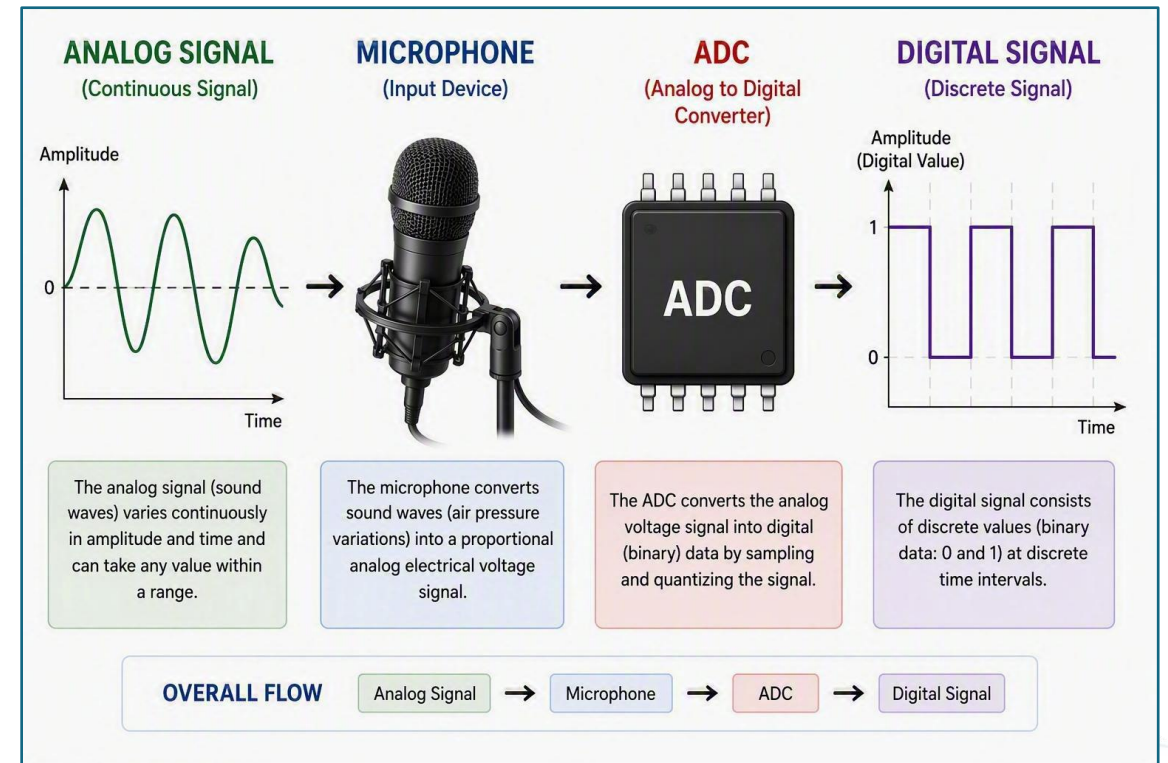


DIGITAL AUDIO RECORDING

ADC converts microphones analog sound signals into digital form for storage and processing.

Examples

- Smartphones
- Voice recorders
- Music recording systems

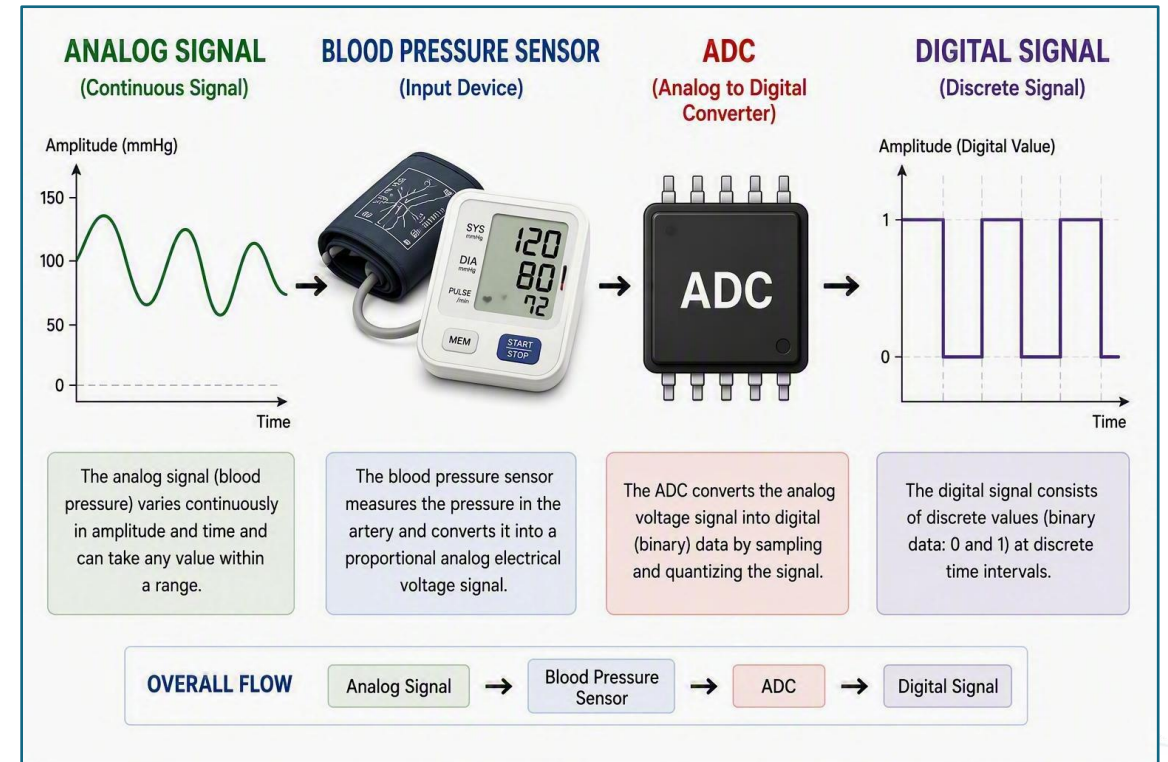


MEDICAL EQUIPMENT

ADC converts biological signals into digital data.

Examples

- ECG (Electrocardiogram)
- EEG (Brain wave monitoring)
- Digital blood pressure monitors

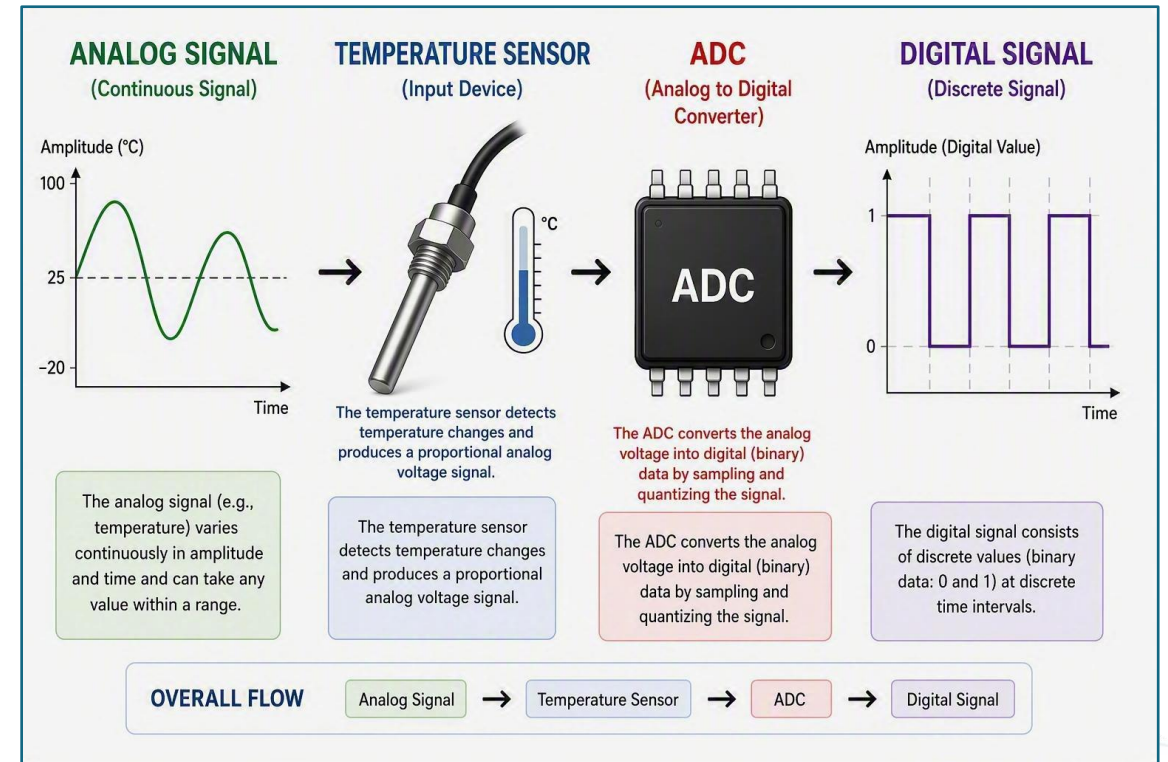


DATA ACQUISITION SYSTEMS

ADC converts measured physical quantities and send them to computers.

Examples

- Temperature monitoring
- Pressure monitoring
- Industrial automation

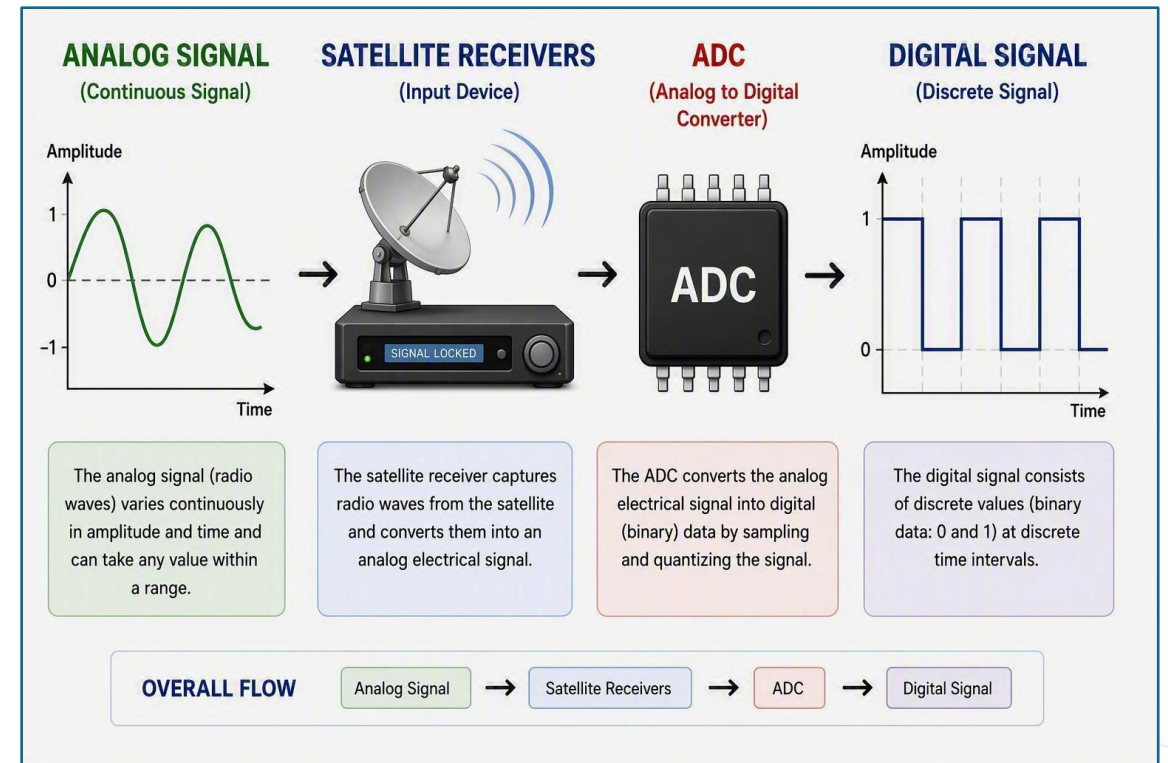


DIGITAL COMMUNICATION SYSTEMS

ADC converts received analog signals into digital data for processing.

Examples

- Mobile phones
- Wireless communication systems
- Satellite receivers



INSTRUMENTATION AND MEASUREMENT

Digital voltmeters and oscilloscopes use ADCs.

ADC enables accurate digital display and analysis of signals.

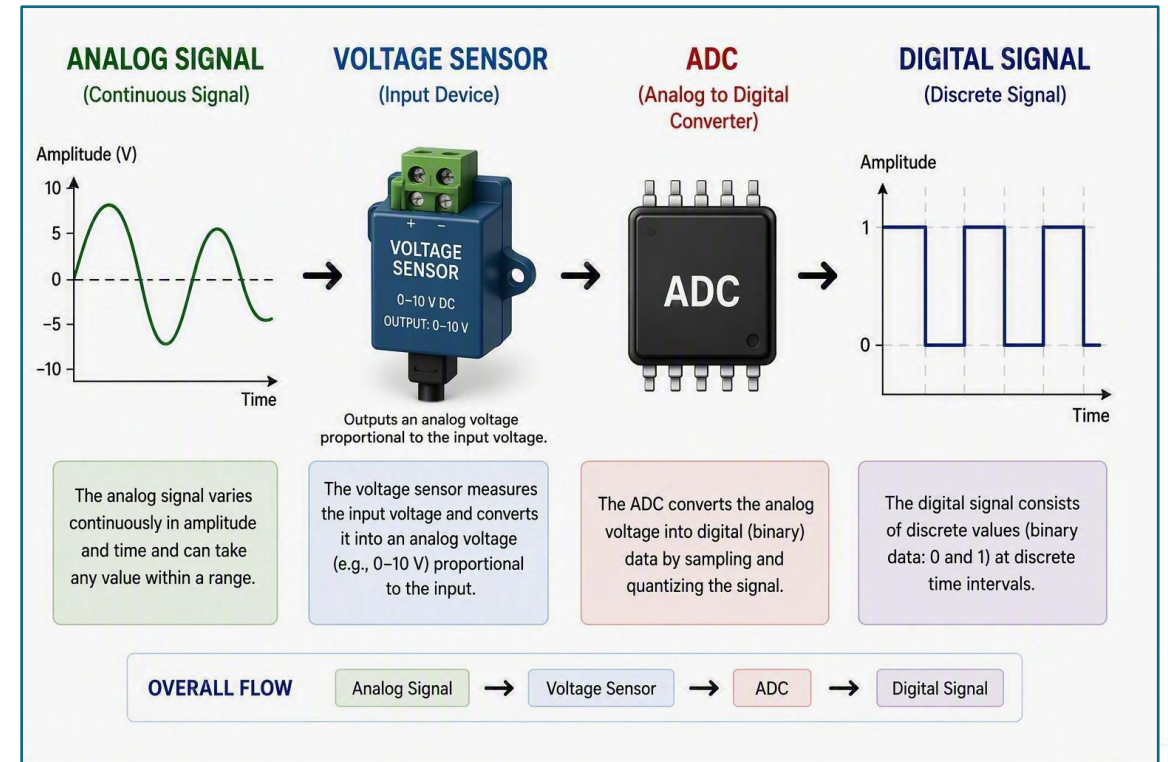
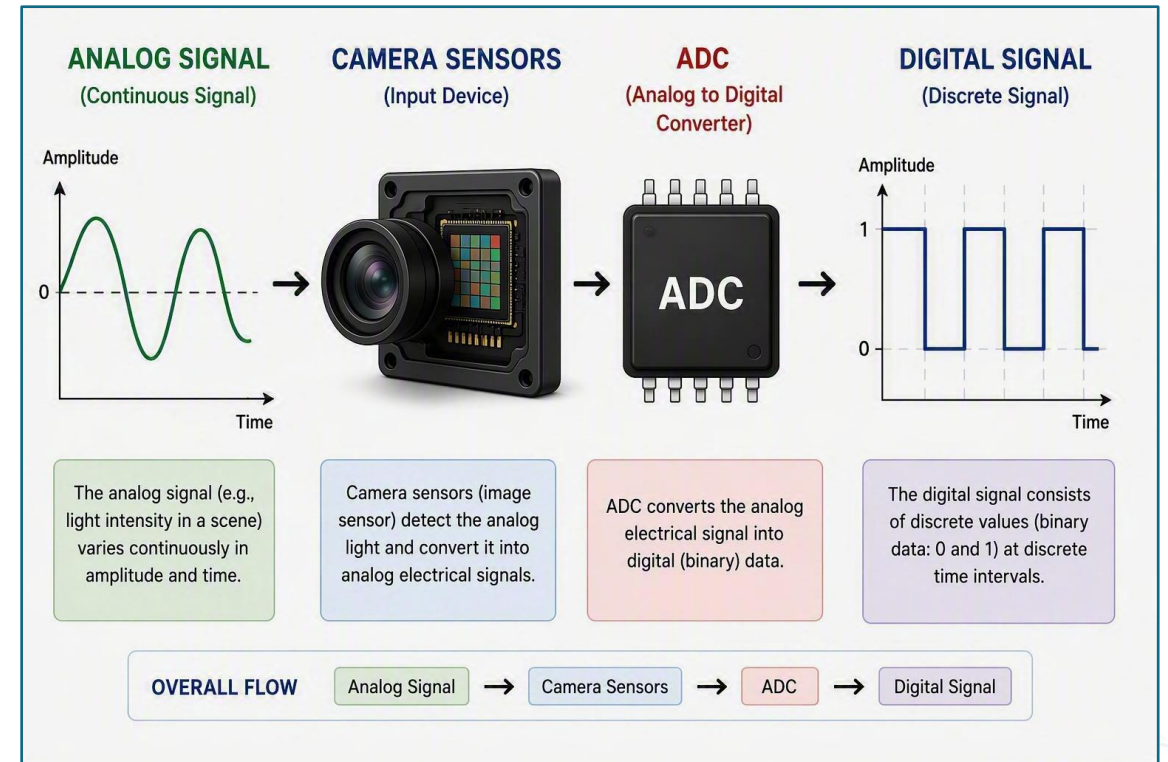


IMAGE AND VIDEO PROCESSING

Camera sensors generate analog signals. ADC converts them into digital images.

Examples

- Digital cameras
- CCTV systems
- Smartphone cameras



DAC



DAC PROCESS

Digital Value

Digital Value = 3102

Decoding

For a 12-bit DAC,

$$\text{DAC} = \frac{\text{Digital Value}}{2^n - 1} \cdot v_{ref}$$

$$\text{DAC} = \frac{3102}{4095} \cdot 3.3$$

Analog Voltage

DAC = 2.5

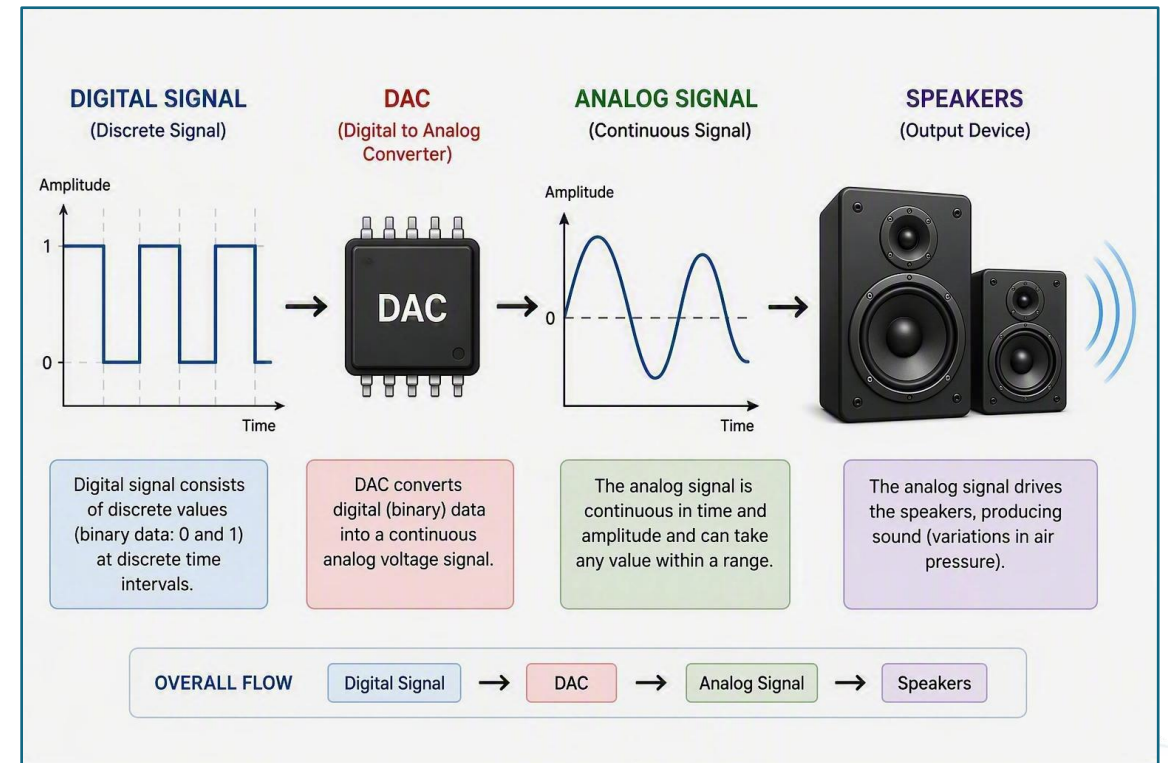


AUDIO PLAYBACK SYSTEMS

DAC converts digital audio files into analog signals for speakers or headphones.

Examples

- MP3 players
- Smartphones
- Computers

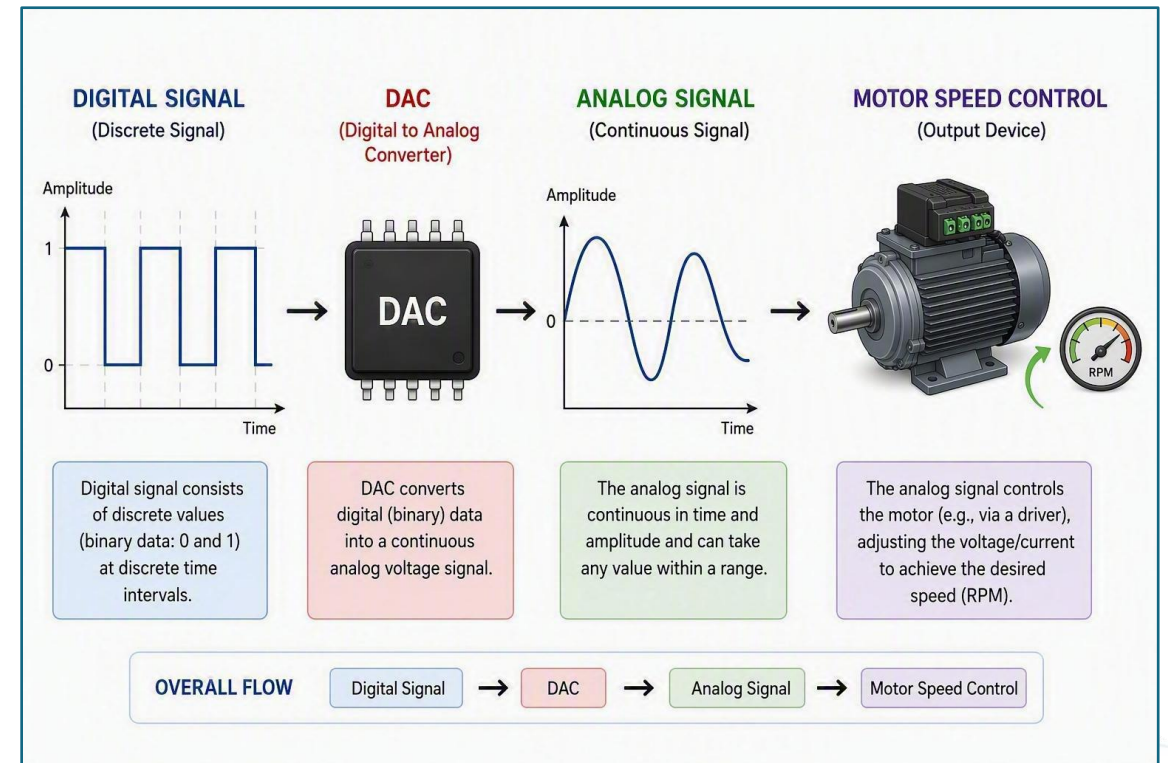


PROCESS CONTROL SYSTEMS

DAC generates analog control signals from digital controllers.

Examples

- Motor speed control
- Industrial automation
- Robotic systems

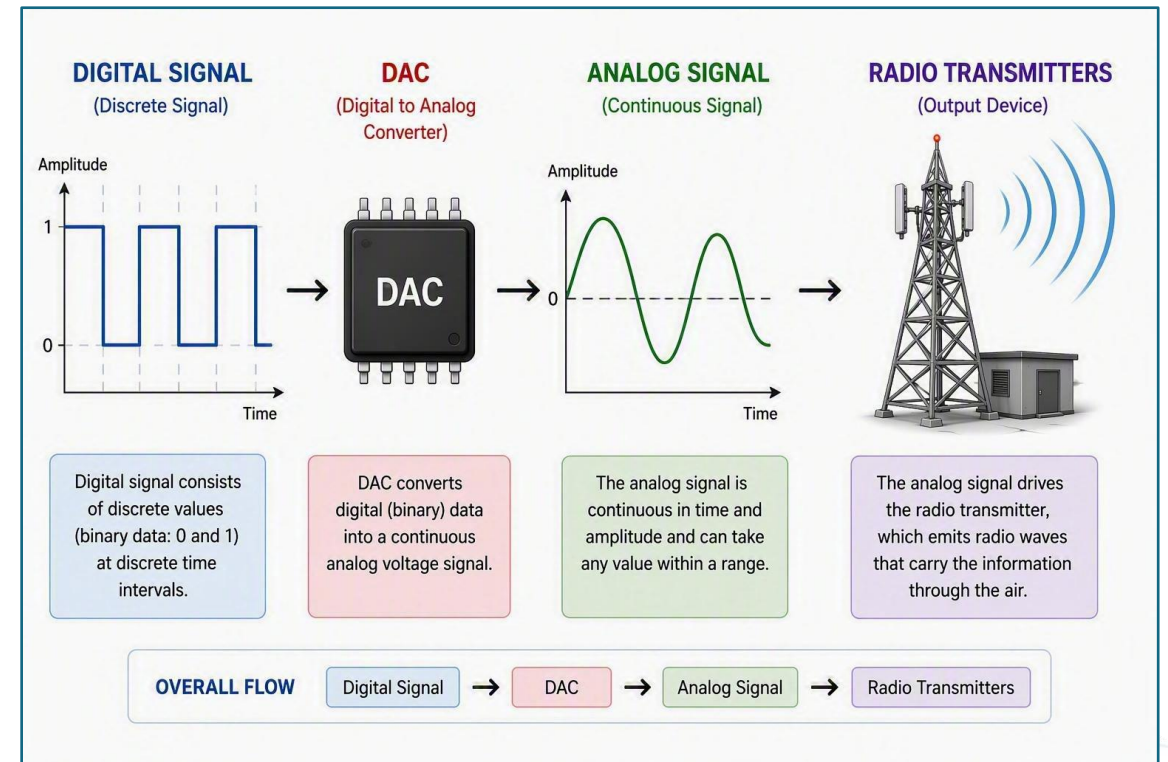


COMMUNICATION SYSTEMS

DAC produces analog waveforms for transmission.

Examples

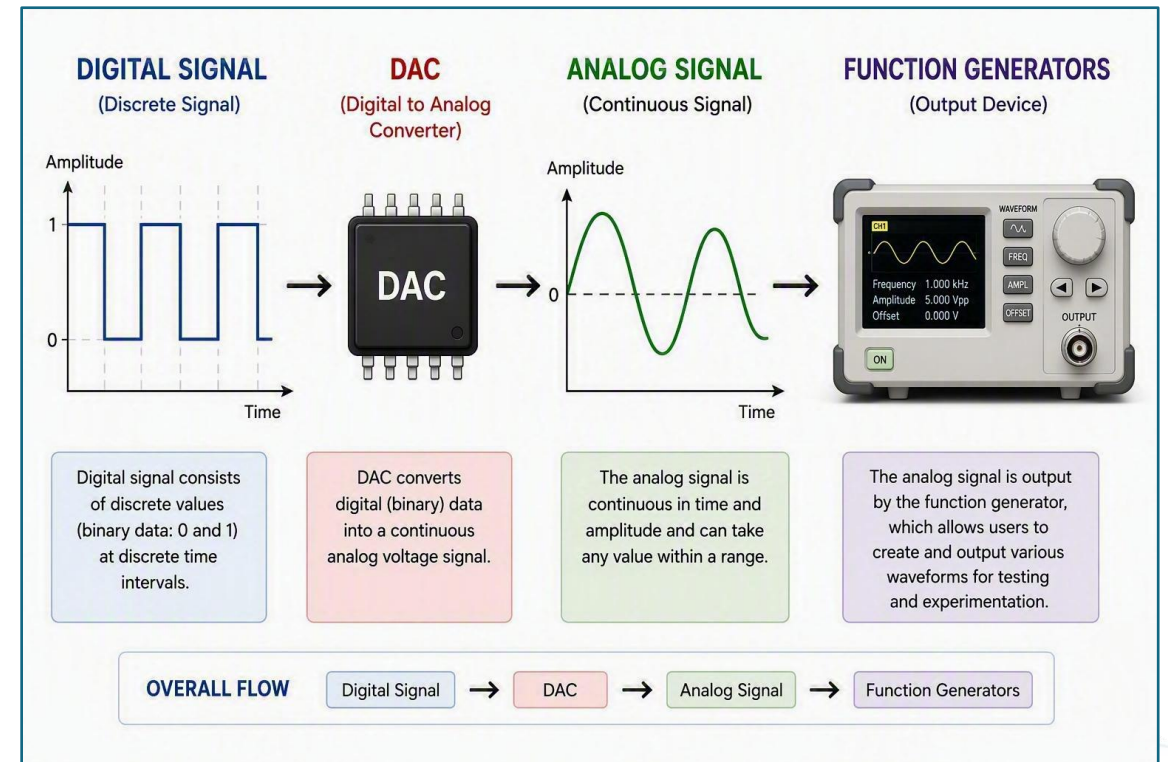
- Radio transmitters
- Modems
- Signal generators



FUNCTION AND SIGNAL GENERATORS

DACs create analog waveforms such as:

- Sine waves
- Square waves
- Triangular waves

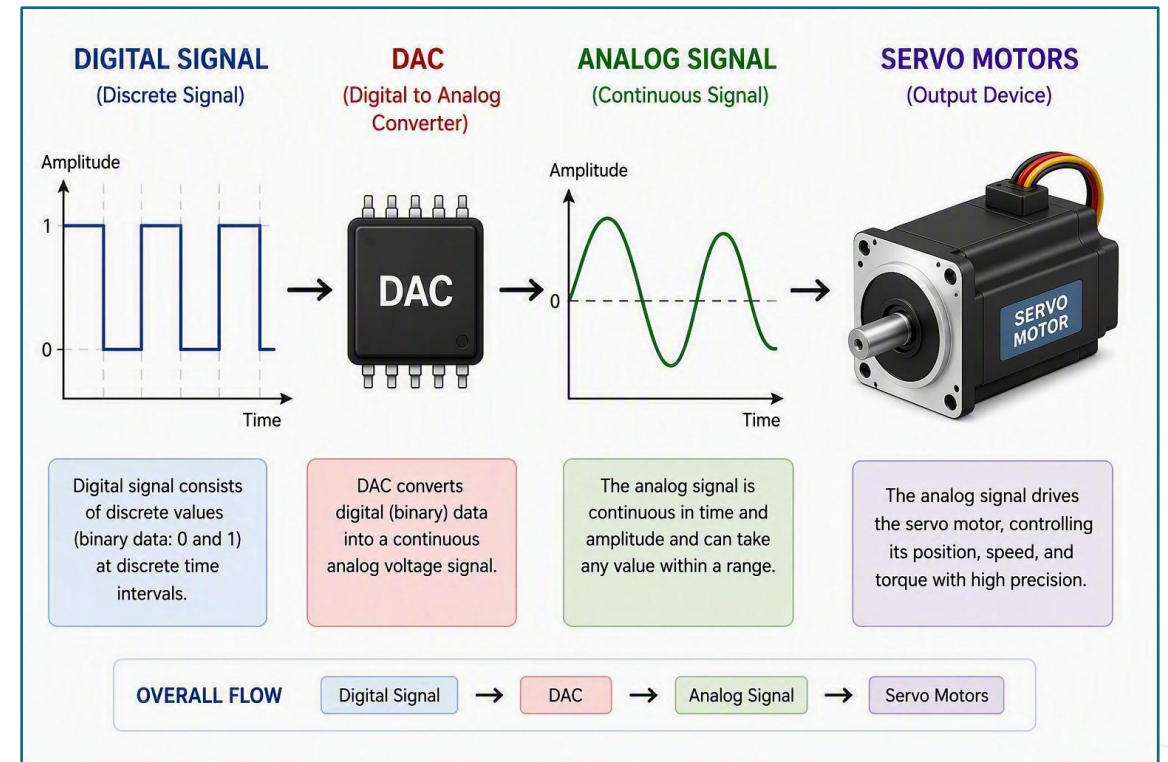


ACTUATOR CONTROL

Digital systems use DACs to control analog devices.

Examples

- Servo motors
- Valves
- Analog meters



THANK YOU

